

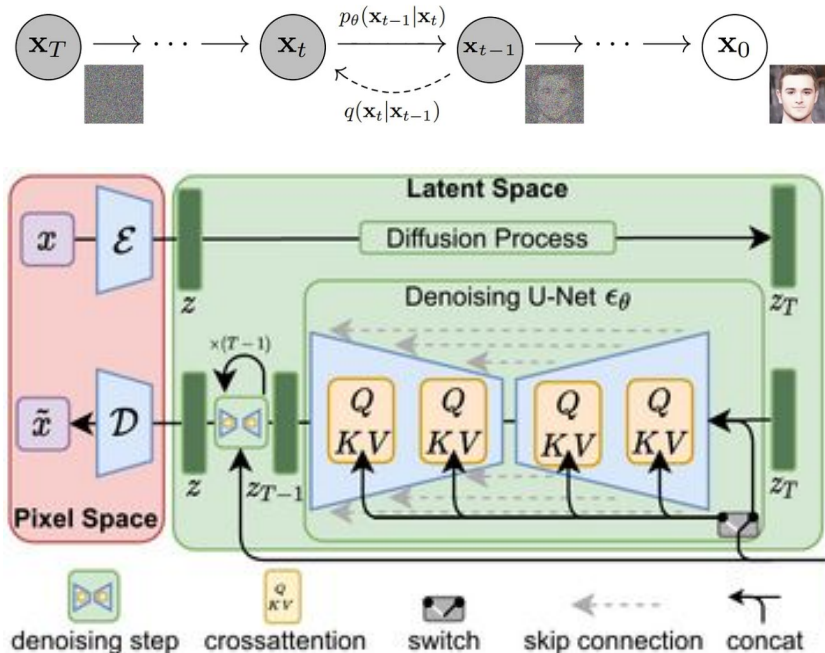
HW5

Unconditional generation

Spring 2026 GAI

Introduction - Diffusion model

- **Forward Process:**
將圖片加入 Gaussian noise 推向 normal distribution.
- **Reverse Process:**
訓練模型學習 denoising, 重建出影像.
- **Latent diffusion model:**
Perform diffusion process in latent space



Task - Professor face image generation

dataset:

- 從NTU/NYCU/NCCU 學術匯聚平台 (Scholar Hubs)爬的教授圖片約 5500張
- 經過人臉裁切、對齊前處理



ref:

<https://scholar.nycu.edu.tw/>

<https://scholars.lib.ntu.edu.tw/browse/rpname>

<https://ah.lib.nccu.edu.tw/browse-scholar-list?page=0&locale=en>

Files

- **images/**: Folder include 5500 professor face training images.
- **train.py**: Sample code to use the training data, and the legal pretrained model.
 - 請參照train.py裡面所使用的pretrain model, 其他的pretrained model都禁止使用.
- **inference.py**: Sample code to generate data.
- **requirements.txt**: Packages used in the following sample code.
- **scoring_program/**: 計算分數的工具與資料

For more detail, please refere to the comments in the above files. 以上的檔案只是範例, 你可以任意修改。

Implementation Rules

- Allowed Models: 允許使用Pretrained VAE 來進行latent space與pixel space的轉換。
- Forbidden Models: 嚴禁使用任何其他預訓練的生成模型(例如:預訓練的U-Net、CLIP 等)。
- Additional data: 可自行從其他學校的學術網站爬取並處理額外的臉部圖像。(使用額外資料可能會導致與提供的資料有分布落差, 請斟酌使用)

Codabench competetion

- 計分方式為Codabench網站競賽 [\[連結\]](#)
- 用訓練的模型生成3,000 張圖像並壓縮上傳至codabench計分。
- All image files must be in PNG format and in resolution 256*256.
- **Deadline: 2025/06/09 23:55 p.m.**
- **資料上傳跟算分需要時間, 請不要到最後一刻才上傳結果**
- 禁止上傳包含任何訓練資料的結果, 被發現直接0分計算
- 請參照投影片後面的Codabench使用說明以及Codabench網站上的說明

Measuring Performance

FID(Fréchet Inception Distance)

- 是目前公認的品質評估指標，專門用來測量生成圖片與真實照片的相似度。
- Extracts features using a pre-trained Inception v3 network.
- Compares the Mean and Covariance of real vs. generated distributions.
- Lower is Better: A lower score indicates generated images are more realistic and diverse.

Grading Policy

Codabench 評分 (100%)

Evaluation Score	Points
FID > 90 (weak baseline)	0
FID < 90 (weak baseline)	70
FID < 50 (strong baseline)	80

分數排名 (Ranking)	Bonus
75% ~ 50%	+5
50% ~ 25%	+10
Top 25%	+15
Top 10%	+20

E3 Submissions

- **Deadline: 2025/06/09 23:59 p.m.**
- Put all the file into the same folder, and submit your zipped file as hw5_{student_id}.zip to E3. After unzip the file, it should appear a folder hw5_{student_id}.
- 繳交範例:
\$unzip hw5_111222333.zip (unzip完之後出現以下檔案)
 hw5_111222333/
 ├── code/ # Your training and inference code
 ├── model/ # Your model weight(如果檔案太大可以給雲端連結)
 └── results/ # Generated 3000 images
- 請按照以上格式繳交, 未按照格式(每一項)會直接扣總分5分。
- 請確保助教可以透過你提供的程式與模型得到結果

Notes

- No plagiarism!!!
- 如果有任何問題請寄信給作業五助教
[TA] 曾薔 chiang.ee11@nycu.edu.tw
- [TA hour] 5/27 (Wed.) 13:00~15:00 at ED716
(Please make an appointment through email first.)

Hint

- Train diffusion model: https://huggingface.co/docs/diffusers/tutorials/basic_training
- Scheduler: <https://huggingface.co/docs/diffusers/api/schedulers/overview>
- Data preprocess: <https://github.com/mantasu/face-crop-plus>

Codabench使用説明

確認E3上登記的信箱



[焦點綜覽](#) / [關於我](#)



電機博 / GEE 曾薔

使用者的詳細資料

[編修個人資料](#)

電子郵件信箱

chiang.ee11@nycu.edu.tw (其他課程的同學也可以看到)

英文姓名

其它

[討論區文章](#)

[討論區議題](#)

註冊帳號

Search Competitions

Benchmarks Resources Queue Management Login Sign-up

Sign Up

帳號為你的學號

使用e3上登記的信箱

☐ I accept the [terms and conditions](#).

[Sign Up](#)

Already have an account? [Log In](#) instead.

註冊帳號

Email收取驗證信以啟用帳號

Activate your user account. 收件匣 ×



Yi-Lun

寄給 我 ▾

5月1

Hi ,

Please click on the link below to confirm your registration:

<http://example.com/profiles/activate/Mg/75b-3f049f05941f5018431f>

This is an automatic message delivered by Codabench.

參加比賽

1. Register for the competition

Get Started

Phases

My Submissions

Results

Forum

?

You have not yet registered for this competition.

To participate in this competition, you must accept its specific [terms and conditions](#). This competition **does not** require approval, once you register, you will immediately be able to participate.

☐ I accept the terms and conditions of the competition.

Register

參加比賽

2. Download the training data, mu and sigma for calculating FID, and scoring program(Sample code to evaluate your results).

Get Started

Phases

My Submissions

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Forum

?

Submission

Scoring Program

Terms

Files

Download	Phase	Task	Type	Size
Image Quality	Development Phase	HW5 FID Evaluation	Scoring Program	4.5 kB
HW5 Reference [Public]	Development Phase	HW5 FID Evaluation	Input Data	32.1 MB
HW5 Public Data	Development Phase	-	Public Data	481.7 MB

參加比賽

3. Submission upload

[Get Started](#)[Phases](#)[My Submissions](#)[Results](#)[Forum](#)[?](#)

Development Phase

[?](#)Number of submissions used for the day

0 out of 10

Number of total submissions used

0 out of 100

Submission upload

Submit as: [?](#)

Yourself



參加比賽

4. There is instruction for scoring program on the website, your can follow the instruction to validate generation by yourself.

The screenshot shows a web application interface. At the top, there is a navigation bar with the following items: 'Get Started' (underlined), 'Phases', 'My Submissions', 'Results', 'Forum', and a circular help icon with a question mark. On the left side, there is a sidebar menu with the following items: 'Submission', 'Scoring Program' (highlighted with a dark background), 'Terms', and 'Files'. The main content area is titled 'Instructions for Running the Scoring Program'. Below the title, there is a paragraph: 'The official scoring program is provided for local evaluation. This is an optional step designed to assist your development process.' Below this paragraph, there is a section titled '1. Download Files' with the text: 'Download the following files from the **Files** tab on the left:'. This is followed by a bulleted list: 'hw6_scoring_program.zip from the **HW5&6 Scoring Program** button' and 'hw6_public_ref.zip from the **HW6 Public Reference** button'. Below the list, there is a section titled '2. Unzip Files' with the text: 'First, unzip hw6_scoring_program.zip into the scoring_program directory:'. Below this text, there is a terminal code block containing the command: 'unzip hw6_scoring_program.zip -d scoring_program'. At the bottom, there is another line of text: 'Then, unzip hw6_public_ref.zip into scoring_program/input/ref:'.

Get Started Phases My Submissions Results Forum ?

Submission

Scoring Program

Terms

Files

Instructions for Running the Scoring Program

The official scoring program is provided for local evaluation. This is an optional step designed to assist your development process.

1. Download Files

Download the following files from the **Files** tab on the left:

- hw6_scoring_program.zip from the **HW5&6 Scoring Program** button
- hw6_public_ref.zip from the **HW6 Public Reference** button

2. Unzip Files

First, unzip hw6_scoring_program.zip into the scoring_program directory:

```
unzip hw6_scoring_program.zip -d scoring_program
```

Then, unzip hw6_public_ref.zip into scoring_program/input/ref:

參加比賽

5. Make sure to select the correct submission to the leaderboard.

Search... 		Status 			
ID # ▾	File name	Date	Status	Score	Add to Leaderboard
3	results2.zip	2026-05-19 00:02	Finished	66.32	  
2	results2.zip	2026-05-18 23:57	Cancelled		
1	results.zip	2026-05-18 20:50	Finished	203.41	 

注意事項

如果你的帳號名稱不是學號，請創建一個organization並以你的學號命名

Get Started

Phases

My Submissions

Results

Forum

Development Phase

?

Number of submissions used for the day

3 out of 3

Number of total submissions used

3 out of 100

Submission upload

Submit as: ?

+ Add New Organization

Yourself

+ Add New Organization

Codabench注意事項

1. 只有用e3上登記的信箱註冊的帳號才能參加比賽，如果e3上登記的信箱無法使用請寄信給助教更換信箱
2. 帳號請直接用學號註冊，帳號刪掉無法用同個email重新註冊，如果帳號(學號)打錯的話，請使用你的學號創建“organization”
3. 請再次確認你在leaderboard上顯示的名稱為學號，如有錯誤會扣分
4. 如果網站與帳號有遇到任何問題請寄信給作業五助教